Dhiraj M. Patil

Machine Learning Engineer — LLM Systems — AI Infrastructure — GenAI
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Summary

Machine Learning Engineer focused on LLM systems, transformer training, and production GenAI applications. Built projects spanning pretraining data pipelines, decoder-only transformer development, retrieval-augmented generation, and cloud deployment. Comfortable across JAX/Flax, PyTorch, Hugging Face, FastAPI, and GCP.

Technical Skills

Languages:	Python, Java, SQL
ML / DL:	JAX, Flax, Optax, Orbax, PyTorch, TensorFlow, Hugging Face Transformers, Hugging Face Datasets, LangChain
LLM Systems:	Transformer Architectures, LLM Pretraining, Fine-tuning, Tokenization, RAG, FAISS, Sentence Transformers
Backend:	FastAPI, Flask, REST APIs, SQLAlchemy
Cloud / DevOps:	GCP, Docker, CI/CD, GitHub Actions, Vercel
Tools:	Git, Linux, Dataset Pipelines, Model Optimization, Distributed Training

Projects

LaughLM Mar 2026 – Present
JAX, Flax, Transformer Architecture — GitHub

- Training a decoder-only transformer LLM from scratch with a modular architecture designed for experimentation and extensibility.
- Implemented modern training components including RoPE positional embeddings and RMSNorm to improve training stability.
- Built tokenization, batching, and preprocessing pipelines for large-scale text datasets used in LLM pretraining.

Text Curation Pipeline Jan 2026 – Present
Python, Hugging Face Datasets — GitHub

- Built reusable data curation workflows for LLM pretraining and fine-tuning.
- Improved preprocessing throughput by **3**× using multiprocessing-based batch transformations.
- Designed scalable dataset transformation modules to support repeatable ML data processing.

AI Chat Platform Oct 2025 – Dec 2025
FastAPI, Hugging Face Transformers, LangChain, GCP — GitHub

- Built a production-style LLM application for conversational AI, document Q&A, and web-assisted responses.
- Implemented Retrieval-Augmented Generation pipelines using FAISS and Sentence Transformers.
- Deployed GPU-backed inference on GCP with a Next.js frontend on Vercel, delivering 1–3s response latency.

Open Source Contributions

Hugging Face Ecosystem

- Initiated a design discussion on reproducibility in Daggr workflows involving external model and Space dependencies across execution environments.
- Proposed dependency provenance tracking using commit hashes and revisions to improve reproducibility, traceability, and auditability.
- Contribution led to implementation of dependency tracking features by maintainers, including merged pull requests (PRs #65, #68).

Education

University of Mumbai Mumbai, India
Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence and Machine Learning) 2025
CGPA: 8.38 / 10

Certifications

- **AWS Data Engineering** (AWS Certified)
- **Generative AI with AWS** (Coursera)